

FRANK PORTER
2016-2017

M1 MACRO

Group B

Session 12

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Permanent Income model:

$$C_t = \alpha A_t + \bar{Y} + \frac{\alpha}{1+\alpha-\rho} \tilde{Y}_t$$

Impulse response function with $\tilde{Y}_{t+1} = \rho \tilde{Y}_t + \varepsilon_t^Y$

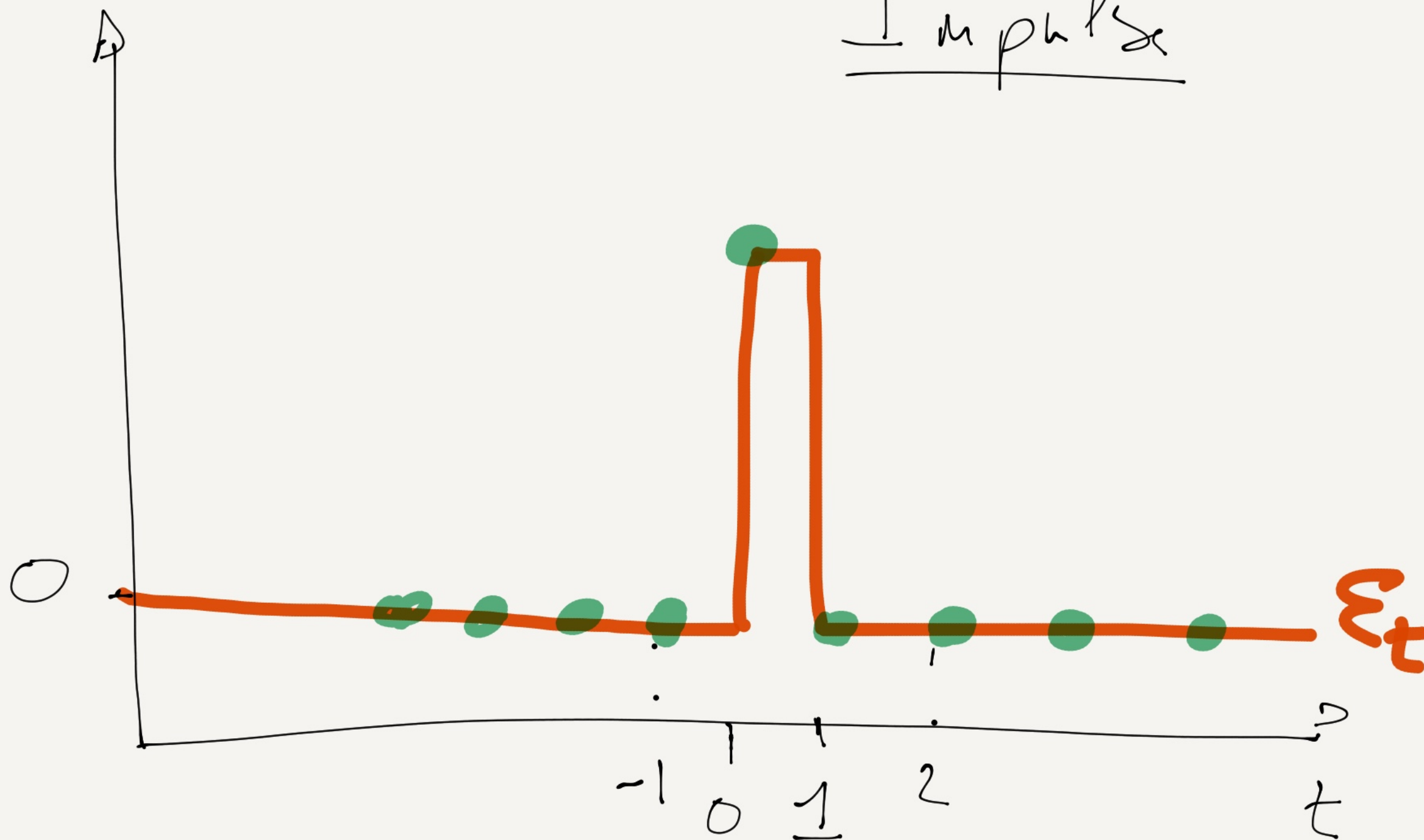
• Assume $\varepsilon_t^Y = 0 \quad \forall t < 0$

so that $\tilde{Y}_{-1} = 0$

$$\left[\tilde{Y}_{-1} = \sum_{i=1}^{\infty} \rho^i \varepsilon_{-i}^Y \right]$$

• Assume that $\varepsilon_0^Y = 1$, $\varepsilon_t^Y = 0 \quad \forall t > 0$

Impulse

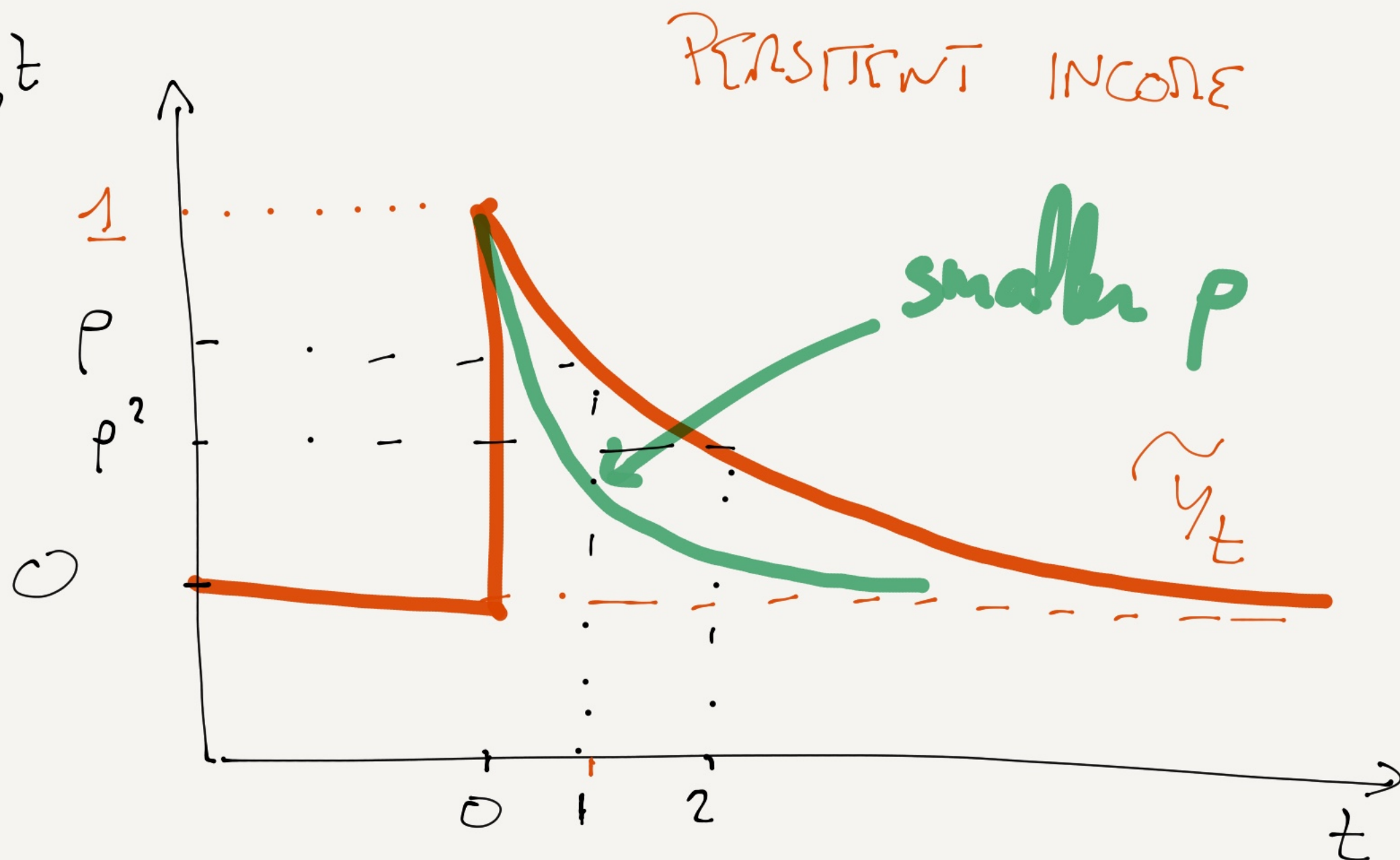


$$\tilde{y}_0 = \rho \tilde{y}_{-1} + \varepsilon_0 = \rho \times 0 + 1 = 1$$

$$\tilde{y}_1 = \rho \tilde{y}_0 + \varepsilon_1 = \rho \times 1 + 0 = \rho$$

$$\vdots$$

$$\tilde{y}_t = \rho^t$$



Consumption

$$C_t = r A_t + \bar{y} + \frac{r}{1+r-p} \tilde{y}_t$$

$$E_t(C_{t+i}) = C_t \quad \forall i$$

$$A_{t+1} - A_t = A \quad \forall t$$

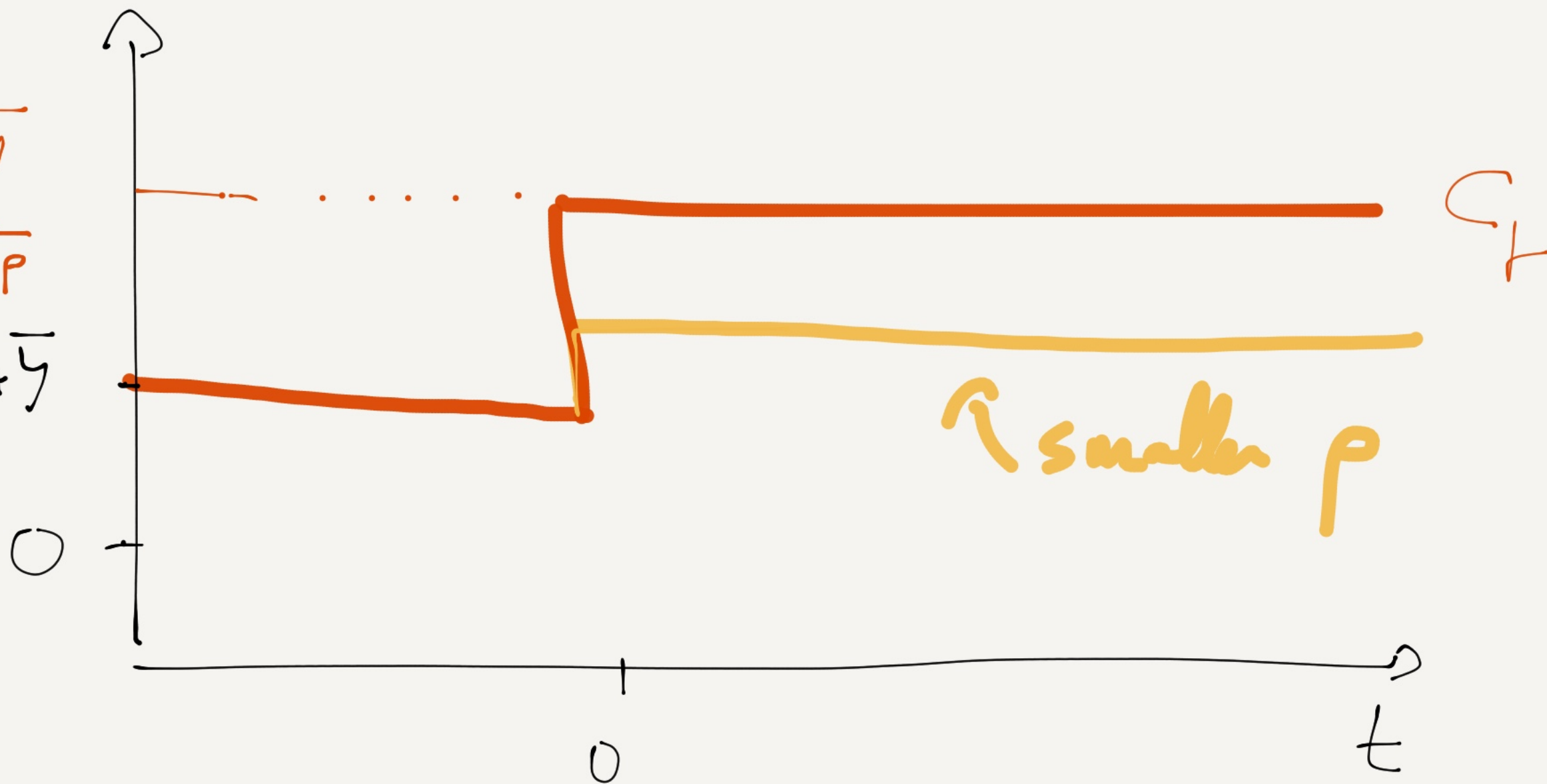
$$C_{-1} = r A + \bar{y}$$

$$C_0 = r A + \bar{y} + \frac{r}{1+r-p} \tilde{y}_0 = r A + \bar{y} + \frac{r}{1+r-p}$$

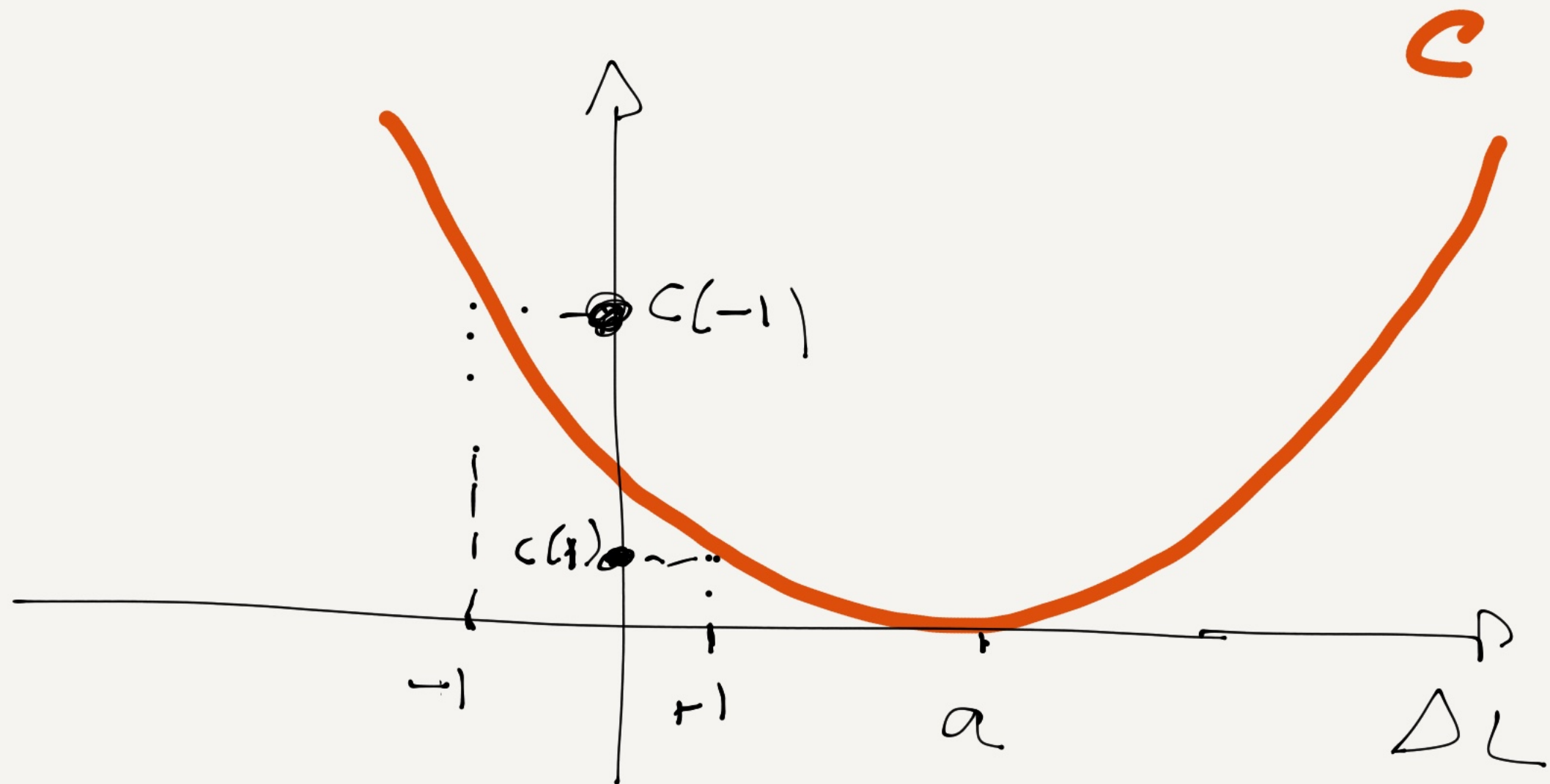
→ No shocks after period 0 → $E_t(C_t) = C_t \quad \forall t > 0$
 $= C_0$

$$\frac{A}{n+1} + \frac{n}{1+n-p}$$

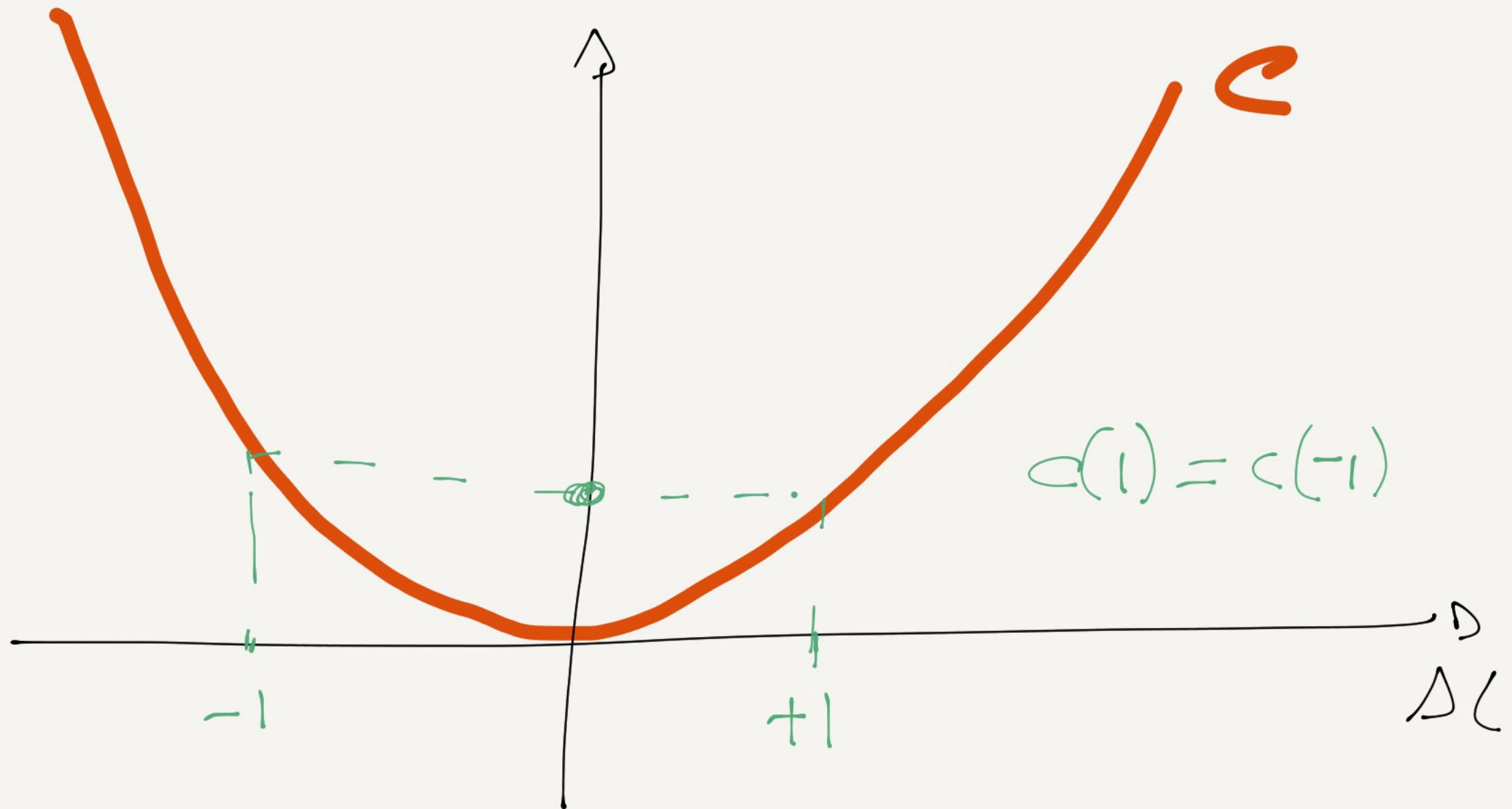
$$nA + \bar{y}$$



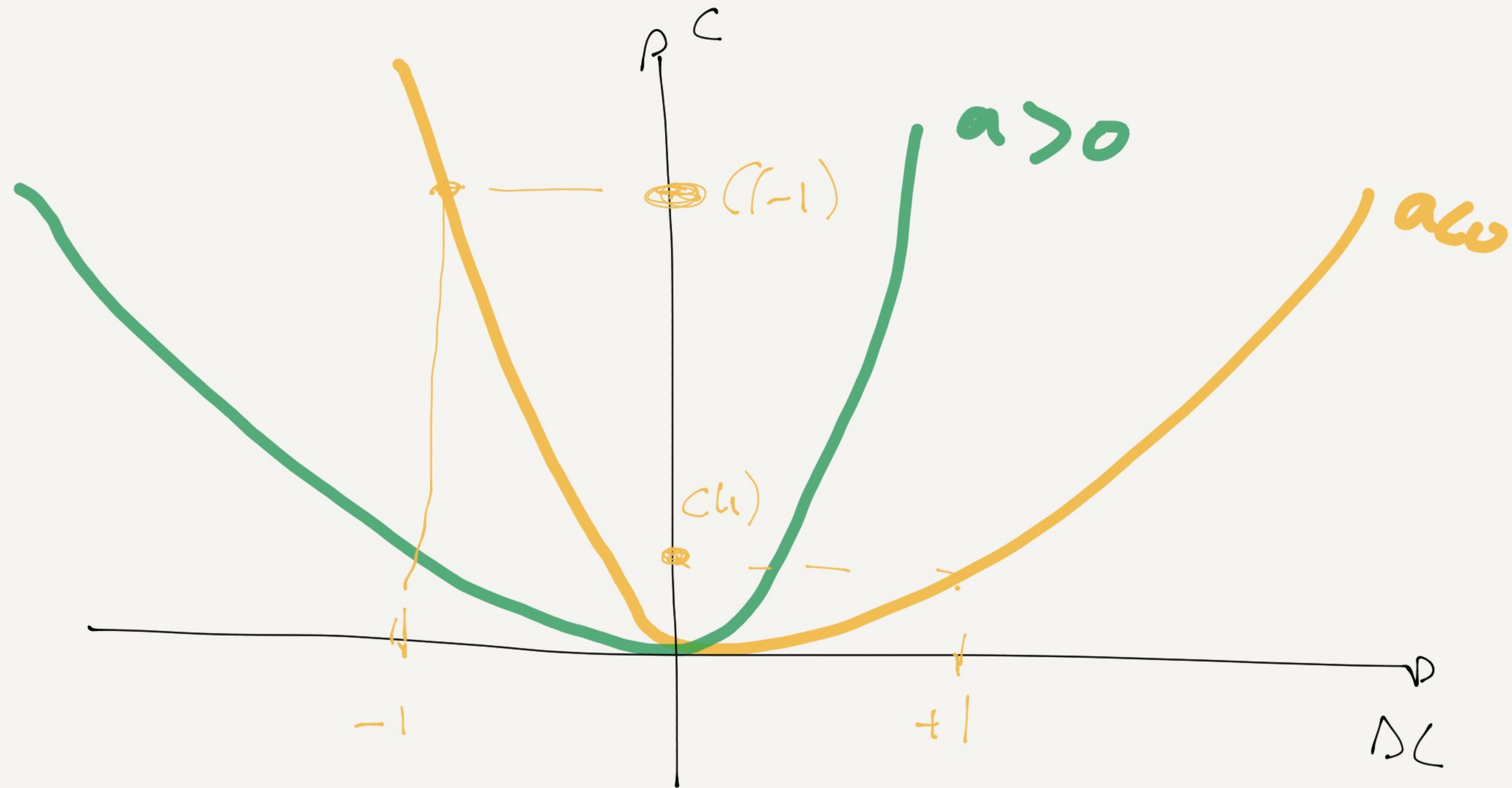
$$C(\Delta L) = \frac{b}{2} (\Delta L - a)^2$$



$$C(\Delta y) = \frac{b}{2} (\Delta L)^2$$



$$C(\Delta L) = -1 + e^{a\Delta L} + a\Delta L + b(\Delta L)^2$$



$$C(\Delta L) = \begin{cases} c_F + b_F \Delta L & \text{if } \Delta L < 0 \\ c_H + b_H \Delta L & \text{if } \Delta L > 0 \end{cases}$$

